Does Marriage Cause Higher Wages for Men?

Does Religion Affect the Returns to Marriage for Men?

Student ID: 1605710

April 2019

Word Count: 4999 (Including tables and equations)

Abstract

This paper discusses the effect of marriage on male wages using contemporary British panel data. Marriage theoretically increases male wages as it induces specialisation within the household. Further, marriage also acts as a signalling device, signalling qualities employers value. Initially using pooled OLS and then a difference in difference estimation, this paper finds a statistically significant increase in male wages following marriage, although this only appears larger for the non-religious due to selection in time invariant characteristics that positively predict both the probability of marriage and wages.

Acknowledgements

The author would like to thank Dr Ariana Ornaghi for her many helpful contributions that made this dissertation possible. Also, thanks to my family, who are a constant source of help. Cheers.

Table of Contents

Section 1: Introduction 3

Section 2: Literature Review 5

2.1 Overview and Theory 5

2.2 Empirical Approaches 5

2.3 Contribution 7

Section 3: Data 8

Section 3.1 Dataset 8

Section 3.2 Key Variables 8

Section 3.3 Restrictions 9

Section 3.4 Limitations of the Data 10

Section 3.5 Descriptive Statistics 10

Section 4: Methodology 11

4.1 Outline 11

4.2 Pooled OLS 11

4.3 Difference in Difference 12

Section 5: Analysis 14

5.1 Pooled OLS 14

5.2 Difference in Difference 16

Section 6: Supporting Evidence 17

6.1 Overview 17

6.2 Lag and Lead Model 18

6.3 Logit 19

Section 7: Robustness Checks 20

7.1 Robustness to sample selection 20

7.2 Robustness to religion intensity 21

Section 8: Limitations 21

Section 9: Conclusions 22

References 24

Appendix 26

Section 1: Introduction

The decline of marriage has been a feature of society across the developed world in the last 50 years (OECD, 2018). In the UK, the marriage rate has declined by 48% between 1970 and 2016.

Figure 1: Marriage rates over time, selected OECD countries.

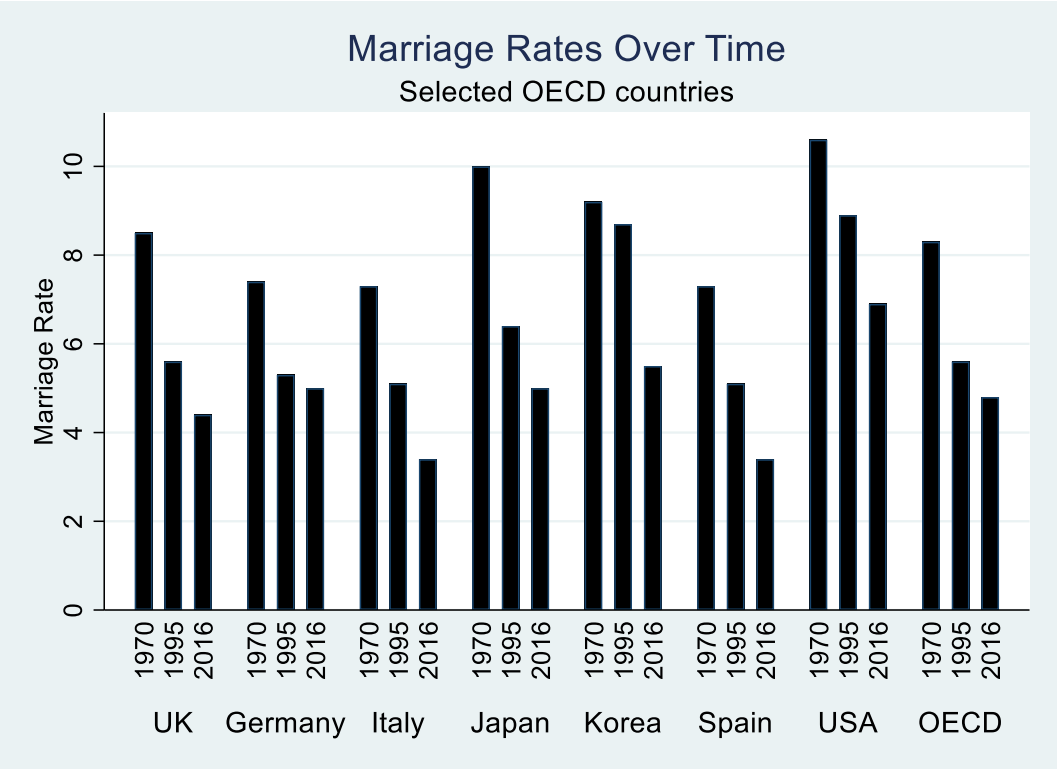

Figure 1 (OECD, 2018), marriage rate is the number of marriages per 1000 people per year.

This is considered an adverse trend by some Conservative politicians, who legislated minor tax cuts for married couples in the UK in 2014 (Finance Act, 2014). Marriage is encouraged by conservatives partially because researchers frequently find positive effects associated with marriage from improved levels of happiness (Blanchflower and Oswald, 2004) and reported health (Oswald and Wilson, 2005). Additionally, marriage has positive effects on wages, named "the marriage wage premium" (Korenman and Neumark, 1991). This usually attributed to either specialisation within the household (Becker, 1991**)** or signalling effects **(**Bardasi and

Taylor, 2005). However, this relationship is not necessarily causal, as perhaps individuals who get married have other favourable characteristics that are rewarded in the labour market, or that individuals with high wage growth are those who get married (Korenman and Neumark, 1991). There are issues of selection bias and reverse causality.

Religion could influence the size of the marriage wage premium. It is possible that the premium may be larger for the non-religious as it causes additional specialisation. Some specialisation may have occurred before marriage in the religious population as this group could be argued to be more traditional (Morgan, 1987).

The UK is an interesting case study as most research focusses on the USA (Cheng, 2016). This research seeks to be innovative and addresses whether this relationship exists in the UK using more recent data, as it is possible the marriage rate decline has resulted in a change in the returns to marriage compared to older research.

This research focusses exclusively on the effect of marriage on male wages. The effect of marriage on female wages is interesting but requires a very different analysis as many females leave the labour market after marriage in order to raise children.

The three main objectives of this study are:

1. To investigate the marriage-wage relationship using UK data from 2009 to 2018. This extends the analysis on UK data substantially, as the most recent study on UK data used data from 2003 as the most recent year analysed (Bardasi and Taylor, 2008).
2. To test whether the income marriage relationship differs between religious and non-religious groups.
3. Address the issue of selection bias and reverse causality in the marriage-wage relationship.

Section 2: Literature Review

2.1 Overview and Theory

The study of marriage has been extensive across economics. Marriage has been found to increase happiness (Blanchflower and Oswald, 2004), reported health (Oswald and Wilson, 2005) and wages (Korenman and Neumark, 1991). The difference in earnings between married and single men is called the "marriage wage premium" (Korenman and Neumark, 1991) and is positive and statistically significant for men.

The primary reason that marriage has a positive effect on male wages is outlined by Becker (1991). He argues that marriage facilitates specialisation in the family, with a male specialising in labour market related tasks and the female in domestic tasks. The male wage subsequently increases because of specialisation and higher productivity. Additionally, marriage signals higher productivity to employers, as marriage is associated with unobservable characteristics that are prized highly in the labour market such as ability and dependability (Bardasi and Taylor, 2005). Other mechanisms by which marriage causes higher wages include employer discrimination, motivational and maturing effects from marriage (Ribar, 2004).

2.2 Empirical Approaches

Early papers quantifying the effect of marriage on wages utilise cross-sectional data and OLS regressions. Schoeni (1995) uses cross sectional data across fourteen developed countries using data from the early 1980s to demonstrate a substantial male marriage premium by regressing wages on marital status and several potential control variables. He finds that in half of the countries, marriage increases male wages by over 20% compared to being single (Schoeni, 1995, p.357). Loh (1996) used OLS on US data and found a smaller premium; married men earn 9.1% more than single men holding observable characteristics constant. Overall, cross

sectional evidence shows married men earn approximately 10-40% more than single men holding observable factors constant (Korenman and Neumark, 1991).

The major limitation of OLS cross-sectional analysis is the difficulty in assigning causality to marriage due to selection bias. The issue of selection reflects that individuals with desirable skills such as strong social skills are both more likely to get married and more likely to earn higher wages (Killewald and Gough, 2013, p.3). Failure to control for these variables will bias the marriage coefficient in an upward direction.

The typical way to address this issue of selection in previous studies was to use difference in difference models. This allows researchers to control for selection into marriage as a fixed effect estimator controls for all time invariant individual level characteristics, such as social skills (Killewald and Gough, 2013). This eliminates the selection problem assuming that the confounding variables are time invariant. Results show the male marriage wage premium is significantly reduced after estimating individual and time level fixed effects, falling from 11% to 6% (Korenman and Neumark, 1991). Approximately one half of the marriage premium is due to selection, but there remains a significant, causal effect of marriage on wages after accounting for selection. The identifying assumption in these papers is that married male wages would have followed the trend of unmarried male wages if they did not marry. Conditional on this assumption, marriage causes a 6% increase in male wages. The limitation is that these papers fail to address issues of time variant confounding variables and reverse causality (Killewald and Lundberg, 2017). However, they improve substantially on the OLS specification by partially addressing selection bias.

More recent papers question whether marriage causes higher wage growth, arguing instead that higher wage growth results in a higher probability of marriage (Killewald and Lundberg, 2017). Cheng (2016) finds evidence of higher wage growth in the marriage group before marriage,

but this growth is exacerbated post marriage, suggesting a causal effect of marriage on wages. If there is evidence of reverse causality, one can utilise instrumental variables to strip out the endogenous variation in the marriage variable. Bardasi and Taylor (2005) implement this method and find no evidence that reverse causality was generating bias.

2.3 Contribution

Whilst the current literature is extensive, most researchers interested in the marriage wage relationship analyse older US data. Only Bardasi and Taylor (2005, 2008) analyse the UK using data from 2003 as the most recent year analysed. As social norms around marriage are changing, results may be different from prior studies. This thesis addresses whether the premium has changed in the subsequent two decades.

Furthermore, very few papers add interactions (the notable exception being gender interactions), explaining what may change the size of the marriage premium for different individuals. This is surprising as there is considerable heterogeneity in the marriage wage premium for men across countries (Schoeni, 1995), so it is pertinent to address why the effect of marriage differs across individuals.

Religion may reduce the magnitude of the marriage wage premium via reducing the effect of specialisation. Religious women may already somewhat specialised in household tasks and men in labour tasks, being more traditional than their non-religious counterparts. This may dampen the specialisation effect that Becker (1991) argues is the mechanism that causes higher wages for men.

Section 3: Data

Section 3.1 Dataset

This paper uses data exclusively from *Understanding Society* (USoc) (University of Essex, 2018). USoc is a longitudinal dataset led by the Institute for Social and Economic Research (ISER). The dataset contains yearly data from the same individuals on a plethora of socioeconomic variables, with over 1400 variables available in the first wave. The raw dataset contains 373,775 observations through all 8 waves from 84,925 unique individuals. The datasets panel nature is crucial as this dataset tracks the same individuals every year, measuring changes in wages, and marital status. This allows analysis that utilises changes in wage trajectory after marriage, in particular use of a difference in difference research design that allows causal inference regarding the effect of marriage on wages.

Section 3.2 Key Variables

The dependant variable is the natural logarithm of gross monthly labour income (measured in nominal pounds). The variable reflecting marital status has five categories: single; never married; cohabiting; married/ civil partnership; divorced/separated and not cohabiting; divorced/separated and now cohabiting. These are mutually exclusive and exhaustive. For instance, if an individual is married, they are no longer deemed cohabiting. Religion is a dummy variable (1 if religious, 2 otherwise), determined by asking if an individual holds a religious belief.

Beyond this, in all estimating equations there is a vector of control variables that are used to keep constant other factors that affect income. Table 1A in the appendix shows a complete list of variables.

Section 3.3 Restrictions

Restrictions are imposed on the data due to the nature of the research question and limitations of the data.

Analysis is restricted to employees in the labour force, that is 18 to 65. Inactive and unemployed individuals have no wage and therefore the effect of marriage on wages cannot be estimated. The self-employed are not analysed as there are issues of missing wage data, and the self-employed systematically understating their income for tax reasons (Hurst, Li and Pugsley, 2014).

Analysis is restricted to males because a fundamentally different model is required to accurately assess the impact of marriage on female wages, as many females leave the labour market after marriage in order to raise children.

Individuals who fail to respond in at least two subsequent waves of the survey are dropped. For difference in difference estimation, two subsequent observations are required from each individual in order to estimate fixed effects. Individuals with missing marital status and religion are also dropped as the analysis is impossible to perform if this data is missing.

Individuals who change religion over the dataset are dropped, as comparing marriage returns for religious and non-religious individuals becomes difficult to define if individuals change religion.

Finally, those who earn small wages are dropped. This is necessary as some individuals report negative earnings or earnings considerably below minimum wage. Consequentially, individuals earning less than 300 pounds per month, the individuals in the bottom one percentile of earnings amongst the employed population, are dropped.

To summarise restrictions on the data:

Variable	Restriction
Sex	Male only
Income	Income>£300.00 per month only
Age	18≤Age≤65 only
Employment Status	Full time or Part time only
Number of time observations	T≥2
Religious	Time invariant

After restrictions, 35,532 observations remain from 7,818 males.

Section 3.4 Limitations of the dataset

The main limitation of the dataset is attrition. Attrition arises when individuals leave the dataset. As shown in table 2A, there is an over 65% attrition rate between waves 1 and 8. This results in sample selection bias if individuals that have positive wage outcomes after marriage are more likely to stay in the sample (Miller and Wright, 1995). There is evidence that higher income individuals are more likely to continue responding in certain subsamples (Lynn et al, 2012), suggesting plausible upward sample selection bias.

Another minor limitation is the censored dependent variable. Most individuals earning over £8333 gross pounds per month have income rounded down to this amount. Approximately 1.5% of all observations are censored. Such censoring results in upward bias (Rigobon and Stoker, 2007), but as the proportion of observations is small, the bias is minimal.

Section 3.5 Descriptive Statistics

Table 3A and 4A show the mean of monthly labour earnings across marital status for religious and non-religious males. For religious males, the difference in mean wages between

married and single groups is 40.1%. For non-religious males, the difference is larger at 53.2%. This suggests that marriage has a positive effect on wages that is larger for the non-religious. However, a large portion of this difference is typically not due to marriage but other variables that positively predict entry into marriage and higher wages such as the participant having a university degree. The analysis therefore controls for variables that influence both marriage and wages to directly estimate the effect of marriage. Table 5A gives selected summary statistics for all variables, including control variables.

Section 4: Methodology

4.1 Outline

This paper utilises two different regression techniques, pooled OLS followed by a difference in difference specification. Pooled OLS is a baseline model and shows a conditional difference in mean earnings between married and single individuals. Difference in difference analysis then identifies a causal relationship subject to a number of assumptions. The methodology builds on work from Bardasi and Taylor (2005) with additions from Angrist and Pischke (2008), as well as innovations of my own, such as testing differences between religious and non-religious individuals and testing the assumptions required for difference in difference analysis to be successful. For all specifications, robust clustered standard errors at the individual level will be used to allow for both heteroscedasticity and errors to be correlated across time periods for a given individual but independent across individuals (Cameron and Miller, 2015).

4.2 Pooled OLS

The pooled OLS regression will take the form:

$$\ln(earnings_{it}) = \alpha + \beta_1 cohabiting_{it} + \beta_2 married_{it} + \beta_3 seperated_{it} + \beta_4 recohabiting_{it} + \gamma X_{it} + c_t + \varepsilon_{it}$$

The dependent variable is the log of gross monthly labour earnings. Cohabiting, married, separated and recohabiting are a series of marital status compared to a default of single, never married and X_{it} (see table 1A) is a vector of control variables and c_t represents time fixed effects. Marital statuses are mutually exclusive and exhaustive, such that the interpretation of the marital status coefficients is always compared to the control group (single, never married). The equation is estimated for both religious and non-religious individuals, and the coefficient of interest is β_2, the difference in wages between married and single individuals holding other variables constant, which is anticipated to be positive and significant for both subgroups, although significantly larger for the non-religious.

Estimating this equation using pooled OLS fails to account for time invariant characteristics that may bias the marriage coefficient (Killewald and Gough, 2013). In particular, variables such as social skills and ambition may be positively correlated with both wages and the probability of getting married, causing upward bias of β_2.

4.3 Difference in Difference

To identify a causal relationship, a difference in differences model leveraging individual fixed effects will then be estimated:

$$\ln(earnings_{it}) = \alpha_i + \beta_1 cohabiting_{it} + \beta_2 married_{it} + \beta_3 seperated_{it} + \beta_4 recohabiting_{it} + \gamma X_{it} + c_t + \varepsilon_{it}$$

Again, this regression will be performed on both religious and non-religious populations. The difference is the estimation of fixed effects, α_i. This addresses the issue of unobservable time invariant individual-level heterogeneity. Adding fixed effects resolves the issue of selection bias in time invariant characteristics because the fixed effects estimator controls for the effect of time invariant characteristics. This reduces the problem of omitted variable bias to unobserved confounding variables that vary over time and across individuals such as attractiveness. Table 1A shows the list of control variables used, which differs from the

pooled OLS estimation as time invariant characteristics are colinear with fixed effects and must be dropped.

The inclusion of fixed effects changes the interpretation of the model to a difference in difference model. Marriage is considered a treatment, and the control group is single individuals. Other marital statuses are control treatments, included to prevent omitted variable bias.

For difference in difference analysis to be valid, it must be that in the absence of treatment, the treatment group would have followed the trend of the control group. If this "parallel trends" assumption holds, the trend of single individuals can be used to estimate a counterfactual for the wages of married individuals had they remained single. The treatment effect β_2 is calculated by contrasting the wage growth rate in the single and married group after treatment. β_2 is the difference in difference estimator and represents the causal change in wages from getting married relative to a counterfactual of the same person remaining single.

For β_2 to reflect a causal, unbiased treatment effect, assumptions are required:

1. There must be parallel trends. The wage growth rate of single and married individuals must be the same before treatment.
    a. A subset of assumption 1 is no reverse causality. There must not be evidence that high wage growth increases the probability of marriage.
2. Time variant unobserved characteristics must be independent of marital status.

Section 5: Analysis

5.1 Pooled OLS

Table 1 shows the pooled OLS specification results. After controlling for observable characteristics such as age and highest qualification, the marriage wage premium falls from 40.1% in the raw data for religious males down to 11.5% ($e^{0.109} - 1$). Much of the raw data premium is explained by selection in observable characteristics because married men have characteristics that are associated with higher wages. For instance, married men are in expectation 11.7 years older than single men in this dataset and this is associated with more on the job experience that results in higher productivity therefore greater wages.

Table 1: The effect of marriage on wages (Pooled OLS)

Variables	(1) Aggregate	(2) Religious	(3) Non-Religious
Cohabiting	0.0898***	0.0500*	0.115***
	(0.0143)	(0.0293)	(0.0169)
Married	0.152***	0.109***	0.183***
	(0.0138)	(0.0213)	(0.0183)
Separated	0.0747***	0.00983	0.121***
	(0.0219)	(0.0328)	(0.0288)
Recohabiting	0.122***	0.0428	0.167***
	(0.0247)	(0.0437)	(0.0303)
Constant	6.142***	5.291***	6.958***
	(0.432)	(0.154)	(0.121)
Observations	35,532	15,752	19,780
R-squared	0.497	0.524	0.485

Robust standard errors clustered at the individual level in parentheses. Included and not reported are control variables listed in table 1A**.**
*** p<0.01, ** p<0.05, * p<0.1

In specification 1, which regresses on all males, males earn 16.4% more than non-married males holding other characteristics constant. This difference is significantly different from 0 at the 1% significance level. The difference in wages between single and married individuals is larger in the non-religious subgroup. Using specifications 2 and 3, the marital wage premium is 8.6% points larger for the non-religious males, and this is significant at the 5% level. This is

found by testing the interaction term in the regression in table 6A, which is an approximate test for testing the differential effect of marriage between religious and non-religious individuals in table 1. These results demonstrate a positive marriage wage premium for men that is larger for non-religious males.

However, OLS estimation can result in an upwardly biased coefficient on marriage for both the religious and the non-religious due to the issue selection bias; selection into marriage based upon unobservable characteristics that either cannot be measured or were not recorded. For instance, an individual with strong social skills is possibly more likely to marry and more likely to earn higher wages. Omitting such variables leads to upward bias and is impossible to control for using OLS as social skills are not observed in this dataset.

The differential marriage wage premium between religious and non-religious males may also be driven by different intensities of selection based upon unobservable characteristics. In particular, there may be lower selection based upon unobservable characteristics in the religious sample as marriage is a higher frequency event in this sample, and as such, the selection bias may be smaller for the religious group than the non-religious group and this may be driving the differential in the “marriage wage premium”.

5.2 Difference in Difference

To address selection bias, a difference in difference model is estimated. This specification controls for time invariant confounders by adding fixed effects.

Table 2: The effect of marriage on wages (Difference in Difference)

Variables	(1) Aggregate	(2) Religious	(3) Non-Religious
Cohabiting	0.0359***	0.0223	0.0397***
	(0.0119)	(0.0280)	(0.0133)
Married	0.0631***	0.0499**	0.0649***
	(0.0145)	(0.0253)	(0.0179)
Separated	0.0697***	0.0479*	0.0792***
	(0.0190)	(0.0278)	(0.0264)
Recohabiting	0.0316	-0.0341	0.0575**
	(0.0226)	(0.0383)	(0.0284)
Constant	6.723***	6.020***	7.320***
	(0.276)	(0.403)	(0.353)
Observations	35,532	15,752	19,780
R-squared	0.150	0.152	0.153
Number of unique individuals	7,818	3,592	4,226

Robust standard errors clustered at the individual level in parentheses. Included and not reported are fixed effects, time fixed effects and control variables listed in table 1A**.**
*** p<0.01, ** p<0.05, * p<0.1

It appears that much of the premium for religious males is due to selection in time invariant, unobservable characteristics. The estimated effect of marriage falls by 6.4% points to a 5.1% increase in gross monthly wages for religious males from marriage when comparing the results of pooled OLS to difference in difference specifications. For non-religious individuals, this fall is much larger, falling from approximately 20.1% in table 3 using OLS to 6.7% using a difference in difference specification. A test of whether marriage has a higher coefficient for the non-religious yields a p-value of 0.628 (Table 7A). It is not possible to reject the null hypothesis that the treatment effect of marriage is the same for the religious and non-religious, even at the 10% significance level.

The differential effect of marriage on the non-religious was not because marriage has higher returns for the non-religious, but because selection into marriage based upon time invariant unobservable characteristics that positively predict wages was more intense in the non-religious group. For instance, an individual may need to be more ambitious or socially skilled to get married in the non-religious subgroup and this generates additional selection bias in the non-religious OLS regression. This is plausible as marriage is a higher frequency event in the religious subpopulation (74% of religious observations were married compared to 54% of non-religious observations) and this results in less ambitious or socially skilled individuals getting married in the religious group, reducing the selection bias in the religious group.

Overall, this analysis demonstrates that marriage does have a causal increase on male wages. However, this causal increase is equivalent across religious and non-religious sub groups. Any higher returns to marriage for the non-religious that appeared in the raw data and OLS analysis disappeared with the introduction of fixed effects, implying that selection in unobservable characteristics makes it appear that marriage returns are higher for the non-religious rather than a genuinely larger marriage effect for the non religious.

Section 6: Supporting Evidence

6.1 Overview

The causal inference in Section 5.2 is contingent upon assumptions of parallel trends (and within that, no reverse causality) and an assumption that time variant characteristics that are omitted are independent of marriage to get an unbiased estimate of the treatment effect. It is impossible to test whether omitted time variant characteristics are independent of marriage as they are unobservable. It is however possible to provide evidence that the parallel trends assumption is justified before marriage occurs and that there is no evidence of reverse causality.

6.2 Lag and Lead Model

The parallel trends assumption is that in the absence of treatment, the treatment group would have followed the trend of the control group (Angrist and Pischke, 2008). Although impossible to directly test, it is possible to test whether the trends of the treatment and control group are parallel before treatment, testing whether or not wage growth is the same in the married and single groups before marriage takes place. If this is the case, it is argued that the wage growth would have remained parallel if not for marriage. This test can be implemented by constructing a lag and lead model:

$$\ln(earnings_{it}) = \alpha_i + \beta_0 married_{it} + \sum_{\tau=1}^{6} \beta_{-\tau} married_{it-\tau} + \sum_{\tau=1}^{7} \beta_{+\tau} married_{it+\tau} - \beta_1 married_{it+1} + \beta_8 single_{it}$$
$$+\beta_9 cohabiting_{it} + \beta_{10} separated_{it} + \beta_{11} recohabiting_{it} + \gamma X_{it} + c_t + \varepsilon_{it}$$

In the above equation, the lag effects of treatment are represented by $\sum_{\tau=1}^{6} \beta_{-\tau} married_{it-\tau}$. They show the treatment effect each time period after treatment, for instance, $\beta_{-1} married_{it-1}$ shows the treatment effect one period after a marriage formation compared to the treatment effect one period before a marriage formation.

If the treatment effect is significant before treatment actually takes place, in this case greater wage growth in the marriage group compared to the single group before a marriage formation, this indicates non-parallel trends. When the lead effects of treatment are insignificant, this is evidence that the parallel trends assumption is justified because parallel trends exists prior to treatment.

Figure 2: Event Study Graph showing treatment effects over time.

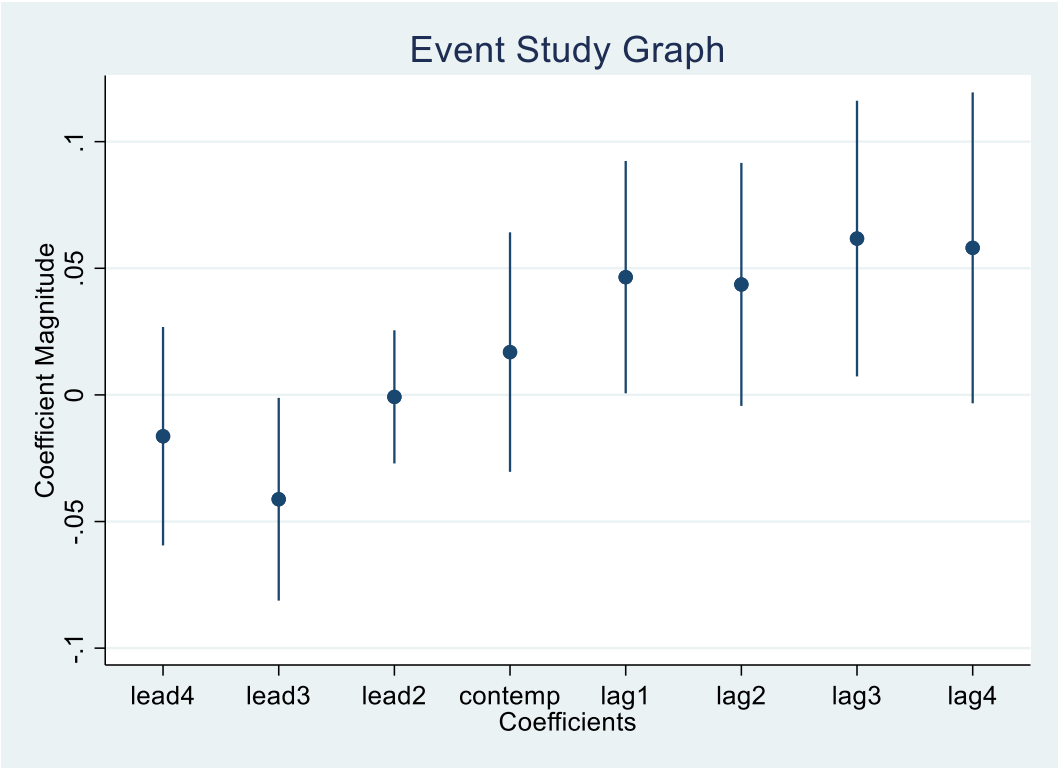

Figure 2 is a coefficient plot of the lag and lead model in table 8A. As the lead effects of treatment are jointly insignificant, this demonstrates that wage growth was broadly equal in the treatment and control groups before marriage, rejecting the hypothesis of reverse causality. It also implies that the parallel trends assumption holds in the pre-treatment period, therefore single individuals are a reasonable counterfactual for married individuals in the absence of marriage. The event study graph also shows the structure of the treatment effect, showing that marriage causes 5% higher wages one year after marriage and remaining constant thereafter.

6.3 Logit

A logit model can also be used to demonstrate parallel trends and a lack of reverse causality.

$$\Pr(Treatprepost_{it} = 1) = \Lambda(\alpha + \beta_1 \Delta \ln(earnings_{it}) + \beta_2 \ln(earnings_{it}) + \gamma X_{it})$$

The dependent variable is the probability of entering the treatment group. This specification is only estimated for those who have never been married when they enter the dataset.

Table 3: Logit model predicting entry into the treatment group.

Variables	(1) Coefficients	(2) Marginal effects at Mean
Cohabiting	1.610***	0.132***
	(0.0969)	(0.0089)
Δln(earnings)	-0.269	-0.018
	(0.170)	(0.0116)
ln(earnings)	0.237**	0.016**
	(0.0999)	(0.0068)
Constant	-6.192***	
	(0.963)	
Observations	7,495	7,495

Robust standard errors in parentheses. Control Variables listed in table 11A.
*** p<0.01, ** p<0.05, * p<0.1

Table 3 shows that growth in wages is negative and bordering on statistical significance at the 10% significant level. This suggests wage growth is lower in the married group before treatment. Using single individuals as a counterfactual for married individuals had they remained single may result in an underestimation of the true treatment effect. The argument that high wage growth is causing marriage is therefore not supported.

Section 7: Robustness Checks

7.1 Robustness to sample selection

This robustness check ensures that the results of the analysis are similar regardless of sample. A balanced sample is a sample where each panel member is observed over the entire sample period. An unbalanced sample arises when individuals are not observed over the entire sample period (Baltagi, 2008). So far, the unbalanced sample was used to increase the number of observations. However, results should be robust to sample choice as it should not be that individuals who leave the dataset are those who generate the result of marriage having

a positive impact on wages. Table 9A shows difference in difference regressions performed on the balanced sample. The result of marriage having a positive effect on wages that is independent of religion is the same regardless of sample used.

7.2 Robustness to religion intensity

This robustness check demonstrates that results are not being influenced by individuals who profess a religious belief but do not actively engage in religious activities. Intensely religious are defined as "intensely religious" if every time an individual is asked they:

i. define themselves as belonging to a religion

ii. answer the question "How much difference would you say religious beliefs make to your life? Would you say they make..." with "a great difference" (ISER, 2009)

Difference in difference regressions are then performed on only "intensely religious" individuals, "somewhat religious" individuals who say that religion makes some to no difference to their daily life, and the non-religious. The results are shown in table 10A. The results demonstrate that marriage returns are slightly lower among the intensely religious, although it is statistically insignificant. This implies that individuals who profess religious belief but are not intensely religious are not influencing the results of the analysis in a significant way.

Section 8: Limitations

The main limitations of the analysis are the assumptions made to perform the analysis. They were of:

1. Parallel trends
2. Independence of marriage to time variant confounders
3. No reverse causality

Section 6 provides supporting evidence that assumptions 1 and 3 are reasonable in this context, however, the assumption that marriage is independent of time variant confounders is a problematic assumption and violating it may bias results in either direction. A particular time variant omitted confounder of concern is a lag of earnings. Including lag of earnings would address potential omitted variable bias and allay concerns caused by earnings being persistent. This is because when the lag of wages is correlated with marriage, omitting it results in bias in the marriage coefficient. The inclusion of lagged dependent variable and additional time variant confounders such as attractiveness would improve the validity of results.

Section 9: Conclusions

The central hypothesis of this paper was that marriage had a positive causal effect on wages that was larger for the non-religious. This paper supports the consensus belief in the literature that marriage has beneficial effects on male wages. It was found to have a similar effect to that found in previous studies but using more recent data in the UK as opposed to most research which takes place in the US using older data. This research finds that marriage increases wages by 6.5% relative to remaining single (table 2 specification 1), and this is a substantial increase. To give a sense of magnitude, the returns to marriage are approximately ¼ of the returns to a degree using OLS estimates from Walker and Zhu (2013).

However, little evidence was found supporting the hypothesis that non-religious males received additional wage benefits of marriage relative to religious males. It only appeared this way due to selection in time invariant unobservable characteristics such as strong social skills. As selection bias was larger in the non-religious subgroup, once this bias was controlled for the effect of marriage was equal for religious and non-religious individuals.

The findings also lend some support to the government policy to give tax breaks to married couples in 2014, as it has a positive effect on male wages and is therefore a wage increasing policy for males if it does increase the probability of marriage. However, as this analysis is limited only to the effect on male wages, there may be adverse effects that offset the beneficial effect on male wages.

Further research could be directed at different interactive terms that may explain the different returns to marriage observed in different countries (see Schoeni, 1995). For instance, research could be conducted on whether marriage has higher returns for the poor or lower returns for females.

References

Angrist, J.D. and Pischke, J.S., 2008. *Mostly harmless econometrics: An empiricist's companion*. Princeton university press.

Baltagi, B., 2008. *Econometric analysis of panel data*. John Wiley & Sons.

Bardasi, E. and Taylor, M.P., 2005. Marriage and wages. *ISER Working Paper Series. No.2005-01*.

Bardasi, E. and Taylor, M.P., 2008. Marriage and wages: A test of the specialization hypothesis. *Economica*, 75(299), pp.569-591.

Becker, G.S,1991. *A treatise on the family*, Cambridge MA: Harvard University Press.

Blanchflower, D.G. and Oswald, A.J., 2004. Well-being over time in Britain and the USA. *Journal of public economics*, 88, pp.1359-1386.

Cameron, A.C. and Miller, D.L., 2015. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, *50*(2), pp.317-372.

Cheng, S., 2016. The accumulation of (dis) advantage: The intersection of gender and race in the long-term wage effect of marriage. *American Sociological Review*, *81*, pp.29-56.

Finance Act 2014, c.26. Available from: http://www.legislation.gov.uk/ukpga/2014/26/contents/enacted [Accessed 25h March 2019]

Hurst, E., Li, G. and Pugsley, B., 2014. Are household surveys like tax forms? Evidence from income underreporting of the self-employed. *Review of economics and statistics*, *96*(1), pp.19-33.

Institute for Social and Economic Research (ISER), 2009. Understanding Society Wave 1 Questionnaire. Available from https://www.understandingsociety.ac.uk/documentation/mainstage/questionnaires [Accessed 25h March 2019]

Killewald, A. and Gough, M., 2013. Does specialization explain marriage penalties and premiums?. *American sociological review*, 78(3), pp.477-502.

Killewald, A. and Lundberg, I., 2017. New evidence against a causal marriage wage premium. *Demography*, *54*(3), pp.1007-1028.

Korenman, S. and Neumark, D., 1991. Does marriage really make men more productive?. *Journal of Human Resources*, 26(2) pp.282-307.

Loh, E.S., 1996. Productivity differences and the marriage wage premium for white males. *Journal of Human Resources*, pp.566-589.

Lynn, P., Burton, J., Kaminska, O., Knies, G. and Nandi, A., 2012. An initial look at non-response and attrition in Understanding Society. *Understanding Society Working Paper Series, Series number 2012-02*. Available from: https://www.understandingsociety.ac.uk/research/publications/520399 [Accessed 18th March 2019]

Miller, R.B. and Wright, D.W., 1995. Detecting and correcting attrition bias in longitudinal family research. *Journal of Marriage and the Family*, pp.921-929.

Morgan, M.Y., 1987. The impact of religion on gender-role attitudes. *Psychology of Women Quarterly*, *11*(3), pp.301-310.

OECD, 2018. (Organisation for Economic Co-operation and Development), *OECD Family Database*, available at http://www.oecd.org/social/family/database [Accessed 5th January 2019]

Oswald, A.J and Wilson, C., 2005. How does marriage affect physical and psychological health? A survey of the longitudinal evidence. *IZA Discussion Paper No. 1619.*

Ribar, D., 2004. What do social scientists know about the benefits of marriage? A review of quantitative methodologies. *IZA Discussion Paper No. 998.*

Rigobon, R. and Stoker, T.M., 2007. Estimation with censored regressors: Basic issues. *International Economic Review*, *48*(4), pp.1441-1467.

Schoeni, R.F., 1995. Marital status and earnings in developed countries. *Journal of population economics*, 8(4), pp.351-359.

University of Essex. Institute for Social and Economic Research, NatCen Social Research, Kantar Public. (2018). *Understanding Society: Waves 1-8, 2009-2017 and Harmonised BHPS: Waves 1-18, 1991-2009*. [data collection]. *11th Edition.* UK Data Service. SN: 6614, http://doi.org/10.5255/UKDA-SN-6614-12

Walker, I. and Zhu, Y., 2013. The impact of university degrees on the lifecycle of earnings: some further analysis.

Appendix

Table 1A: Complete Variable List

Variables	**Definition**
Natural log of earnings**	The natural logarithm of gross monthly labour income measured in nominal pounds per month.
Single	Equal to 1 if single, 0 otherwise.
Cohabiting**	Equal to 1 if cohabiting, 0 otherwise.
Married**	Equal to 1 if married, 0 otherwise.
Separated**	Equal to 1 if divorced/widowed and no longer cohabiting, 0 otherwise.
Recohabiting**	Equal to 1 if divorced/widowed and now cohabiting with a new partner, 0 otherwise.
Religious***	Equal to 1 if any religious belief, 2 otherwise.
Children**	Number of own children in household.
Disabled**	Equal to 1 if long standing illness or disability, 0 otherwise.
Age**	Years of age.
Age Squared*	Years of age squared.
Immigrant*	Equal to 0 if born in the UK, 1 if native, 2 if missing.
Highest Qualification**	Highest Qualification Attained (7 categories)
Ethnicity*	Ethnicity of participant. (9 categories)
Overtime**	Number of hours worked as paid overtime (5 categories)
Hours Worked**	Number of hours normally worked per week (6 categories)
Work Location**	Where the participant works (7 categories)
Region*	Region the participant lives (13 regions)
Father's Occupation*	Occupation of participant's father (11 categories)
Occupation**	Occupation of participant (5 categories)
Tenure**	Tenure of Participant (4 Categories)

*****Included in only the pooled ols regressions

**Included in both pooled OLS and difference in difference regressions

*** Religious only included in specification 1 of the pooled OLS and diff in diff regressions.

Table 2A: Response rates in the subsample used to conduct analysis

	First Wave person was Interviewed								
Wave	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Wave 6	Wave 7	Wave 8	Total
Wave 1	4,284								4,284
Wave 2	4,284	345							4,629
Wave 3	3,251	345	161						3,757
Wave 4	2,711	251	161	1,981					5,104
Wave 5	2,429	197	124	1,981	204				4,935
Wave 6	2,103	168	97	1,683	204	661			4,916
Wave 7	1,854	147	75	1,495	148	661	182		4,562
Wave 8	1,420	110	55	1,092	91	395	182	-	3,345

Table 3A: Mean of monthly labour earnings by marital status for religious men

Marital Status	Mean Earnings	Standard Deviation	N person years
Single Never Married	2007	1103	2169
Cohabiting	2345	1343	867
Married/ Civil Partner	2812	1745	11631
Divorced/Separated	2243	1230	734
Recohabiting	2373	1373	351
Total	2639	1649	15752

Understanding Society 2009-2018. Monthly Earnings nominal.

Table 4A: Average monthly labour earnings by marital status for non-religious men

Marital Status	Mean Earnings	Standard Deviation	N person years
Single Never Married	1914	1126	3926
Cohabiting	2413	1322	3203
Married/ Civil Partner	2933	1625	10710
Divorced/Separated	2552	1438	1117
Recohabiting	2545	1467	824
Total	2609	1526	19780

Understanding Society 2009-2018. Monthly Earnings nominal.

Table 5A: Summary Statistics of Variables.

Variables	**Mean**	**Standard Dev.**	**Minimum**	**Maximum**
Natural log of earnings	7.707	0.587	5.704	9.721
Single	0.173	0.378	0	1
Cohabiting	0.116	0.321	0	1
Married	0.629	0.483	0	1
Separated	0.051	0.219	0	1
Recohabiting	0.031	0.174	0	1
Religious	1.557	0.497	1	2
Children	0.720	1.019	0	6
Disabled	0.224	0.417	0	1
Age	42.522	11.311	18	65
Age Squared	1936.084	962.831	324	4225
Immigrant	0.498	0.771	0	2
Highest Qualification	2.870	2.009	1	10
Ethnicity	1.887	1.852	1	9
Overtime	2.306	0.612	2	5
Hours Worked	3.898	0.760	1	6
Work Location	2.294	0.721	1	6
Region	6.544	3.132	1	12
Father's Occupation	6.659	3.764	1	12
Occupation	1.908	0.947	1	4
Tenure	3.857	1.022	2	5

*This table shows only the sample used in the regressions, after the restrictions in 3.3 have been imposed.

Table 6A: Alternate Specification of Pooled OLS that shows hypothesis tests

Variables	(1) Alternate Spec
Cohabiting	0.0536*
	(0.0289)
Married	0.115***
	(0.0212)
Separated	0.0113
	(0.0327)
Recohabiting	0.0502
	(0.0434)
Cohabiting*Non-Religious	0.0607*
	(0.0334)
Married*Non-Religious	0.0678**
	(0.0280)
Separated*Non-Religious	0.110**
	(0.0436)
Recohabiting*Non-Religious	0.117**
	(0.0529)
Constant	5.339***
	(0.152)
Observations	35,532
R-squared	0.505

Robust standard errors clustered at the individual level in parentheses. Included and not reported control variables listed in table 1A**.** Also included are interaction terms with every in table 1a with the variable non-religious (such as Children*Non-Religious).
*** p<0.01, ** p<0.05, * p<0.1

The test is testing the significance of the coefficient of the interaction Married*Non-religious.

As shown above, the coefficient is significantly different from zero at the 5% significance level.

Table 7A: Alternate Specification of difference in difference that shows hypothesis tests

Variables	(1) Alternate Spec
Cohabiting	0.0223
	(0.0280)
Married	0.0499**
	(0.0252)
Separated	0.0479*
	(0.0278)
Recohabiting	-0.0341
	(0.0383)
Cohabiting*Non-Religious	0.0174
	(0.0310)
Married*Non-Religious	0.0150
	(0.0309)
Separated*Non-Religious	0.0314
	(0.0383)
Recohabiting*Non-Religious	0.0916*
	(0.0476)
Constant	6.744***
	(0.266)
Observations	35,532
Number of Unique Individuals	7,818
R-squared	0.153

Robust standard errors clustered at the individual level in parentheses. Included and not reported are fixed effects, time fixed effects and control variables listed in table 1A**.** Also included are interaction terms with every in table 1a with the variable non-religious (such as Children*Non-Religious)

*** p<0.01, ** p<0.05, * p<0.1

The test is testing the significance of the coefficient of the interaction Married*Non-religious.

As shown above, the coefficient is insignificantly different from zero.

Table8A: Lag and lead model

Variables	(1) Lag and Lead model
Lead 7	-0.185*
	(0.108)
Lead 6	-0.0276
	(0.0402)
Lead 5	-0.0606*
	(0.0335)
Lead 4	-0.0163
	(0.0220)
Lead 3	-0.0412**
	(0.0204)
Lead 2	-0.000782
	(0.0134)
Contemporaneous	0.0169
	(0.0241)
Lag 1	0.0465**
	(0.0234)
Lag 2	0.0436*
	(0.0245)
Lag 3	0.0617**
	(0.0278)
Lag 4	0.0581*
	(0.0313)
Lag 5	0.0712**
	(0.0304)
Lag 6	0.0919**
	(0.0358)
Single	-0.0300
	(0.0214)
Cohabiting	0.00734
	(0.0206)
Separated	0.0230
	(0.0168)
Recohabiting	-0.0216
	(0.0237)
Observations	35,532
Number of unique individuals	7,818
R-squared	0.151

Robust standard errors clustered at the individual level in parentheses. Included and not reported are fixed effects, time fixed effects and control variables listed in table 1A.
*** p<0.01, ** p<0.05, * p<0.1
Default case is Lead 1 i.e treatment effect 1 period prior to marriage.

Table 9A: Difference in Difference results for the balanced sample.

Variables	(1) Aggregate	(2) Religious	(3) Non-Religious
Cohabiting	0.0351*	0.00919	0.0466**
	(0.0187)	(0.0424)	(0.0217)
Married	0.0791***	0.0660*	0.0861***
	(0.0219)	(0.0348)	(0.0282)
Separated	0.0854***	0.0754**	0.0830**
	(0.0252)	(0.0373)	(0.0327)
Recohabiting	0.0541*	-0.0338	0.0877**
	(0.0293)	(0.0563)	(0.0358)
Constant	6.709***	5.676***	8.009***
	(0.769)	(0.909)	(0.516)
Observations	11,360	4,896	6,464
R-squared	0.157	0.170	0.158
Number of unique individuals	1,420	612	808

Robust standard errors clustered at the individual level in parentheses. Included and not reported are fixed effects, time fixed effects and control variables listed in table 1A.
*** p<0.01, ** p<0.05, * p<0.1

Table 10A: Robustness check controlling for religious intensity

Variables	(1) Intensely Religious	(2) Somewhat Religious	(3) Non-Religious
Cohabiting	0.110	0.0178	0.0397***
	(0.0749)	(0.0371)	(0.0133)
Married	0.0494	0.0615	0.0649***
	(0.0428)	(0.0387)	(0.0179)
Separated	0.0318	0.0974**	0.0792***
	(0.0446)	(0.0447)	(0.0264)
Recohabiting	0.0658	-0.0456	0.0575**
	(0.0839)	(0.0554)	(0.0284)
Constant	5.599***	6.208***	7.320***
	(0.760)	(0.564)	(0.353)
Observations	4,717	7,627	19,780
R-squared	0.169	0.141	0.153
Number of unique individuals	1,122	1,752	4,226

Robust standard errors clustered at the individual level in parentheses. Included and not reported are fixed effects, time fixed effects and control variables listed in table1A.
*** p<0.01, ** p<0.05, * p<0.1

*Note that the sample for religious individuals is smaller as the response rate of the question is not 100%.

Table 11A: List of control variables for the logit model

Variables	**Definition**
Natural log of earnings	The natural logarithm of gross monthly labour income measured in nominal pounds per month.
Change in natural log of earnings	The first difference of the natural logarithm of gross monthly labour income measured in nominal pounds per month.
Cohabiting	Equal to 1 if cohabiting, 0 otherwise.
Religious	Equal to 1 if any religious belief, 2 otherwise.
Children	Number of own children in household.
Disabled	Equal to 1 if long standing illness or disability, 0 otherwise.
Immigrant	Equal to 0 if born in the UK, 1 if native, 2 if missing.
Highest Qualification	Highest Qualification Attained (7 categories)
Ethnicity	Ethnicity of participant. (9 categories)
Overtime	Number of hours worked as paid overtime (5 categories)
Hours Worked	Number of hours normally worked per week (6 categories)
Work Location	Where the participant works (7 categories)
Region	Region the participant lives (13 regions)
Father's Occupation	Occupation of participant's father (11 categories)
Occupation	Occupation of participant (5 categories)